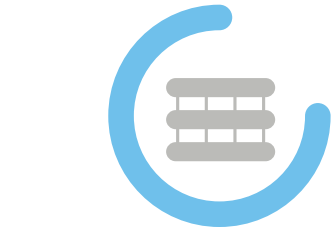


# Meeting of the Technical Steering Committee (TSC) Board

Wednesday, March 11th, 2020
11:00am ET

# Meeting Logistics

- https://zoom.us/j/556149142

- United States : +1  (646) 558-8656
  - Meeting ID: 556 149 142

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# Agenda/Updates

- Announcements:
  - Budget for 2020 Intern program (targeting max of 4 students) has been approved
- Upcoming deadlines:
  - SC'20
    - Tutorials?: Due **April 16, 2020** - *Q: do we want to consider this?*
    - BoFs: Due **July 31, 2020**
- ARM HPC compiler updates
- 2.0 build status
- Continued PEARC'20 tutorial discussion (csim)

# ARM HPC compiler updates

- Have enabled more "arm1" variants for relevant development packages in OBS
  - working through build failures
  - issue created to track failures (https://github.com/openhpc/ohpc/issues/1153)

- scalapack, scotch, and superlu have been updated to build successfully (*Srinath Vadlamani @arm*)
- working on extrae, mumps, plasms, phdf5, sionlib, hypre

| Name | Changed |
|---|---|
| adios-arm1-mpich Link | 2 days ago |
| adios-arm1-openmpi4 Link | 2 days ago |
| arm-compilers-devel | 8 days ago |
| boost-arm1-mpich Link | 15 days ago |
| boost-arm1-openmpi4 Link | 15 days ago |
| dimemas-arm1-mpich Link | 14 days ago |
| dimemas-arm1-openmpi4 Link | 14 days ago |
| extrae-arm1-mpich Link | 14 days ago |
| extrae-arm1-openmpi4 Link | 14 days ago |
| fftw-arm1-mpich Link | 14 days ago |
| fftw-arm1-openmpi4 Link | 14 days ago |
| gsl-arm1 Link | 14 days ago |
| hdf5-arm1 Link | 15 days ago |
| hypre-arm1-mpich Link | 14 days ago |
| hypre-arm1-openmpi4 Link | 14 days ago |
| lmod-defaults-arm1-mpich Link | about 10 hours ago |
| lmod-defaults-arm1-openmpi4 Link | about 10 hours ago |
| metis-arm1 Link | 15 days ago |
| mpich-arm1 Link | 15 days ago |
| mumps-arm1-mpich Link | 14 days ago |
| mumps-arm1-openmpi4 Link | 14 days ago |
| netcdf-arm1-mpich Link | 2 days ago |
| netcdf-arm1-openmpi4 Link | 2 days ago |
| netcdf-cxx-arm1-mpich Link | 2 days ago |
| netcdf-cxx-arm1-openmpi4 Link | 2 days ago |
| netcdf-fortran-arm1-mpich Link | 2 days ago |
| netcdf-fortran-arm1-openmpi4 Link | 2 days ago |
| omb-arm1-mpich Link | 14 days ago |
| omb-arm1-openmpi4 Link | 14 days ago |
| openmpi-arm1 Link | about 1 month ago |
| pdtoolkit-arm1 Link | 6 days ago |
| petsc-arm1-mpich Link | 14 days ago |

# 2.0 Build Status

- Have enabled a good number of packages in 2.0 Factory
  - introduction of arm1 compiler variant means 10 compiler/MPI permutations for MPI-based builds

**Administrative Tools**

| Package | Built? | Notes |
|---|---|---|
| conman | ✓ | |
| docs | ✓ | |
| examples | ✓ | |
| ganglia | | |
| genders | ✓ | |
| lmod-defaults | ✓ | added (3/10/20) |
| losf | ✓ | |
| mrsh | ✓ | fixed (3/8/20) |
| nagios | | |
| nagios-plugins | | |
| ndoutils | | |
| nhc | ✓ | added (3/10/20) |
| nrpe | | |
| pdsh | ✓ | |
| prun | ✓ | |
| test-suite | ✓ | |

**Compiler Families**

| Package | Built? | Notes |
|---|---|---|
| gcc9 | ✓ | |
| intel-compatibility | ✓ | |
| arm-compatibility | ✓ | |
| llvm | | |

**Development Tools**

| Package | Built? | Notes |
|---|---|---|
| easybuild | ✓ | |
| autoconf | ✓ | |
| automake | ✓ | |
| cmake | ✓ | |
| hwloc | ✓ | |
| libtool | ✓ | |
| python-mpi4py | | Adrian built subset ok |
| python-numpy | | Adrian built subset ok |
| python-scipy | | Adrian built subset ok |
| spack | ✓ | [still targets root usage though] |
| valgrind | ✓ | |

**MPI Families**

| Package | Built? | Notes |
|---|---|---|
| impi-compatibility | ✓ | no longer requires install of intel compat. package |
| mpich | ✓ | |
| mvapich2 | ✓ | |
| openmpi4 | ✓ | |

**Serial Libs**

| Package | Built? | Notes |
|---|---|---|
| R | ✓ | |
| GSL | ✓ | |
| metis | ✓ | |
| openblas | ✓ | |
| plasma | X | need tweaks for arm1/intel blas |
| superlu | ✓ | arm1 fixed (3/9/20) |

# 2.0 Build Status (cont.)

**Parallel Libraries**

| Package | Built? | Notes |
|---|---|---|
| boost | X | intel error |
| fftw | ✓ | build fixed |
| hypre | X | intel/arm1 blas issue |
| mfem | X | needs deps |
| mumps | X | arm1/impi issue |
| opencoarrays | X | leap/openmpi4 failures |
| petsc | X | intel/arm1 issue |
| scotch | ✓ | arm1 fixed (3/9/20) |
| scalapack | ✓ | arm1 fixed (3/9/20) |
| slepc | X | waiting on petsc |
| superlu_dist | X | .spec problem (3/11/20) |
| trilinos | | Adrian built subset ok |

**Performance Tools**

| Package | Built? | Notes |
|---|---|---|
| dimemas | X | need some boost builds, suse failures |
| extrae | X | arm1 failures |
| geopm | | Adrian built subset ok |
| imb | | |
| likwid | | |
| msr-safe | | |
| omb | ✓ | 3/4/20 |
| papi | ✓ | |
| paraver | ✓ | added 3/5/20 |
| pdtoolkit | X | intel/leap failure (3/5/20) |
| scalasca | X | need scorep deps (3/5/20) |
| scorep | X | variety of issues (3/5/20) |
| tau | | Adrian built subset ok |

**Resource Management**

| Package | Built? | Notes |
|---|---|---|
| PBS Pro | X | need libical-devel on aarch/centos |
| pmix | ✓ | |
| slurm | ✓ | |

**Runtimes**

| Package | Built? | Notes |
|---|---|---|
| charliecloud | | |
| ocr | | |
| singularity | X | go problems (3/9/20) |

**Provisioning**

| Package | Built? | Notes |
|---|---|---|
| warewulf | ✓ | starting to run in CI (3/10/20) |

# 2.0 Build Status (cont)

- Current package counts:
  - ~800 RPMs as of 3/11/2020

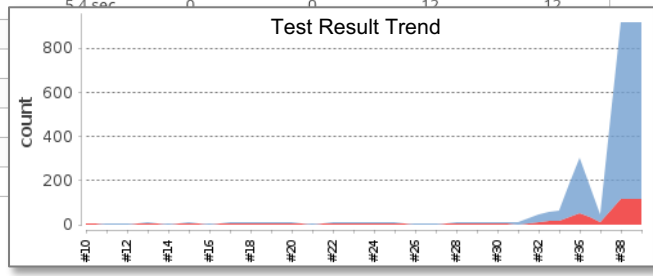
| Base OS | aarch64 | x86_64 | noarch |
|---|---|---|---|
| CentOS 8 | 138 | 244 | 29 |
| Leap 15 | 135 | 233 | 29 |

# 2.0 (cont)

- Mentioned last time that next big-ticket item was provisioning
  - have a basic working WW build now: have been testing x86/centos-8 first
  - variety of issues encountered, pulling from development branch of WW and adding more patches to work with our installation recipes
  - happy to report that basic provisioning of compute nodes is running in a CI job!
- Variety of failures from Jenkins installation test due to missing builds, but jobs are now running in SLURM
  - 610 user-level tests passing
  - expect 822 total for this ethernet only case (so sitting at 74% currently)

**All Tests**

| Class | Duration | Fail | (diff) | Skip | (diff) | Pass | (diff) | Total | (diff) |
|---|---|---|---|---|---|---|---|---|---|
| **Boost** | 32 sec | 0 | | 0 | | 71 | +71 | 71 | +71 |
| **Boost-MPI** | 18 sec | 6 | +6 | 0 | | 20 | +20 | 26 | +26 |
| **Compilers** | 1.9 sec | 0 | | 0 | | 14 | +14 | 14 | +14 |
| **Dimemas** | 9.9 sec | 0 | | 0 | | 14 | +14 | 14 | +14 |
| **Extrae** | 14 sec | 2 | +2 | 0 | | 16 | +16 | 18 | +18 |
| **FFTW** | 12 sec | 1 | +1 | 0 | | 19 | +19 | 20 | +20 |
| **GSL** | 41 sec | 27 | +27 | 0 | | 31 | +31 | 58 | +58 |
| HDF5 | 8.2 sec | 0 | | 0 | | 181 | +71 | 181 | +71 |
| **HWLOC** | 0.58 sec | 0 | | 0 | | 9 | +9 | 9 | +9 |
| **Hypre** | 36 sec | 12 | +12 | 0 | | 30 | +30 | 42 | +42 |
| MPI | 19 sec | 6 | -1 | 0 | | 20 | +1 | 26 | |
| **MUMPS** | 13 sec | 5 | +5 | 0 | | 9 | +9 | 14 | +14 |
| MiniFE | 39 sec | 2 | | 0 | | 8 | | 10 | |
| **NetCDF** | 6.7 sec | 0 | | 0 | | 68 | +68 | 68 | +68 |
| **PAPI** | 21 sec | 0 | | 0 | | 25 | +25 | 25 | +25 |
| **PETSc** | 13 sec | 4 | +4 | 0 | | 20 | +20 | 24 | +24 |
| PHDF5 | 5.7 sec | 2 | | 0 | | 9 | | 11 | |
| **PLASMA** | 2.4 sec | 0 | | 0 | | 10 | +10 | 10 | +10 |
| PNETCDF | 11 sec | 4 | | 0 | | 9 | | 13 | |
| **PTScotch** | 8.5 sec | 3 | +3 | 0 | | 21 | +21 | 24 | +24 |
| **R** | 27 sec | 1 | +1 | 0 | | 7 | +7 | 8 | +8 |
| **SCOTCH** | 0.71 sec | 0 | | 0 | | 12 | +12 | 12 | +12 |
| **SLEPc** | 15 sec | 4 | +4 | 0 | | 18 | +18 | 22 | +22 |
| SUPERLU | 1.1 sec | 0 | | 0 | | 4 | | 4 | |
| **Scalasca** | 20 sec | 3 | +3 | 0 | | 15 | +15 | 18 | +18 |
| **Valgrind** | 1.8 sec | 0 | | 0 | | 10 | +10 | 10 | +10 |
| /apps/hpcg | 3.6 sec | 10 | | 0 | | 4 | | 14 | |
| charliecloud | 0 ms | 5 | | 0 | | 0 | | 5 | |
| easybuild | 6 sec | 0 | | 0 | | 3 | | 3 | |
| **metis** | 2.6 sec | 0 | | 0 | | 13 | +13 | 13 | +13 |
| modules | 5.4 sec | 0 | | 0 | | 12 | | 12 | |
| notest | | | | | | | | | |
| omb | | | | | | | | | |
| openblas | | | | | | | | | |
| **scalapack** | | | | | | | | | |
| **tests** | | | | | | | | | |
| user-env | | | | | | | | | |



# 2.0 (cont)

- Some provisioning gotchas:
  - centos-8 now uses `dnf`
  - can still use `yum`, but all roads lead thru dnf
  - installing into chroot can expose issues with current version of dnf, consider:

```
# yum --installroot /tmp/foo4 install bash
```

```
[BaseOS]
name=CentOS-$releasever - Base
#mirrorlist=http://mirrorlist.centos.org/?release=$releasever&arch=$basearch&repo=BaseOS&infra=$infra
#baseurl=http://mirror.utexas.edu/$contentdir/$releasever/BaseOS/$basearch/os/
baseurl=http://mirror.utexas.edu/centos/8.1.1911/BaseOS/x86_64/os/
gpgcheck=1
enabled=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-centosofficial
```

OK

No Workie

OK

*appears to be a gotcha with dnf version in centos8.1; newer version from Stream works ok with all of the above*

# 2.0 (cont)

- Some provisioning gotchas (cont):
  - previously we owned the build of munge (munge-ohpc)
  - per previous discussions, we decided to rely on distro version instead
  - reintroduces a problem related to consistency of "munge" user/group id creation
  - note that munge creates user/group for itself if not available
    - we need munge on SMS and compute nodes
    - in WW, this means munge is installed into chroot for VNFS assembly
    - no guarantee that munge uid/gid will be the same on host and inside the chroot
    - our recipes suggest synchronizing users across the system via WW file synchronization of {passwd,shadow,groups}
    - the gotcha is that munge cannot start on a compute node because the uid/gid provided from SMS is different than what it created initially in chroot
      - this means SLURM does not work and all our tests go south
  - we dealt with this previously by suggesting a uid/gid in munge scriptlets
  - to work around now, am suggesting to copy passwd/groups from SMS into chroot prior to installing munge:
    - BETTER IDEA: add munge uid/gid creation in ohpc-slurm-client
    - or, create munge-ohpc package that only created uid/gid…

```
# Add OpenHPC components to compute instance
cp /etc/passwd /etc/group  $CHROOT/etc
yum -y --installroot=$CHROOT install ohpc-slurm-client
```

# 2.0 (cont)

- Next up:
  - enable python3 components
  - continuing working through missing builds
  - enable CI for Leap15 and aarch64

- Given how big of a change everything is (particularly on rhel side), starting to think we may want to put out a 2.0 tech preview first to let folks try things out
  - and to perhaps make things available prior to having every single component sorted out….

# PEARC'20 Tutorial Discussion (csim)

- focus on staging ohpc cluster on ec2
- telecon: Tuesday next week (demo of EC2 installation framework)